

Ellipses

ANSWER KEY



Standard form

- 1 For the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$, find:
- a** the vertices $(\pm 5, 0)$
 - b** the foci $(\pm 4, 0)$
 - c** the endpoints of the minor axis $(0, \pm 3)$
 - d** the eccentricity $e = \frac{4}{5}$
- 2 Write the standard-form equation of the ellipse with foci $(0, \pm 3)$ and vertices $(0, \pm 5)$.
Major axis vertical with $a = 5$, $c = 3$, so $b^2 = 25 - 9 = 16$: $\frac{x^2}{16} + \frac{y^2}{25} = 1$.
- 3 Consider the ellipse $9x^2 + 4y^2 = 36$.
- a** Write the equation in standard form. $\frac{x^2}{4} + \frac{y^2}{9} = 1$.
 - b** State the lengths of the major and minor axes. Major axis $2a = 6$ (vertical); minor axis $2b = 4$.
 - c** Find the coordinates of the foci. $c^2 = 9 - 4 = 5$, so foci $(0, \pm \sqrt{5})$.

Applications

- 4 A whispering gallery has an elliptical ceiling 40 m long and 24 m high at the center (so $a = 20$ and $b = 12$, with the major axis horizontal). Two people stand at the foci. How far apart are they, to one decimal place?
 $c = \sqrt{400 - 144} = \sqrt{256} = 16$, so the foci are $2c = 32.0$ m apart.
- 5 An ellipse centered at the origin has a horizontal major axis of length 12 and passes through the point $(3, 2\sqrt{3})$. Find its equation.
 $a = 6$: $\frac{9}{36} + \frac{12}{b^2} = 1$ gives $\frac{12}{b^2} = \frac{3}{4}$, so $b^2 = 16$: $\frac{x^2}{36} + \frac{y^2}{16} = 1$.

Ellipses centered away from the origin

- 6 For the ellipse $\frac{(x-2)^2}{16} + \frac{(y+1)^2}{9} = 1$, find:
- a** the center $(2, -1)$
 - b** the vertices $(2 \pm 4, -1) = (6, -1), (-2, -1)$
 - c** the foci $c = \sqrt{16 - 9} = \sqrt{7}$: $(2 \pm \sqrt{7}, -1)$
- 7 Complete the square to write $4x^2 + y^2 - 8x + 6y + 9 = 0$ in standard form, and state the center.
 $4(x^2 - 2x) + (y^2 + 6y) = -9$: $4(x-1)^2 - 4 + (y+3)^2 - 9 = -9$, so $4(x-1)^2 + (y+3)^2 = 4$, i.e. $(x-1)^2 + \frac{(y+3)^2}{4} = 1$. Center: $(1, -3)$.

Eccentricity and orbital applications

- 8 Compute the eccentricity of each ellipse, and state which is more circular (closer to $e = 0$):

a $\frac{x^2}{100} + \frac{y^2}{36} = 1$ $c = \sqrt{100 - 36} = 8$: $e = \frac{8}{10} = 0.8$ **b** $\frac{x^2}{25} + \frac{y^2}{24} = 1$ $c = \sqrt{25 - 24} = 1$: $e = \frac{1}{5} = 0.2$ —
this one is more circular

- 9 A planet's elliptical orbit has semi-major axis $a = 150$ million km and eccentricity $e = 0.02$. Find the distance from the center to a focus, c , and hence the closest approach distance to the sun ($a - c$), to two decimal places (in millions of km).

$c = ea = 0.02(150) = 3$ million km. Closest approach $= a - c = 150 - 3 = 147.00$ million km.